ANSWER KEY — Integration Lesson 9 Test: Signed Area, Net vs Total

For the grader · full working shown · a bare negative number without a words translation loses half the point on interpretation questions

1. Toy train, $v = -4$ m/s for 3 seconds: compute $\int_0^3 (-4)\,dt$ and interpret.

> **Work:** constant rate: $(-4) \times 3 = -12$. FTC check: $F(t) = -4t$, $F(3) - F(0) = -12 - 0 = -12$ ✓. Below-axis rectangle 3 wide × 4 tall, counted negative. **Answer: −12 → the train ends 12 meters behind its start.**

2. $f(x) = -2x$ on $[0, 3]$: triangle size, then $\int_0^3 (-2x)\,dx$, and why the signs differ.

> **Work:** triangle: base 3, depth 6 (down to -6 at $x = 3$) → size $= \frac{1}{2} \times 3 \times 6 = 9$. FTC: $F(x) = -x^2$ (check: derivative is $-2x$ ✓); $F(3) - F(0) = -9 - 0 = -9$. Signs differ because geometric size is always positive, while the integral counts below-axis regions as negative (they represent losses). **Answer: size 9; integral −9.**

3. Evaluate $\int_0^2 (x - 5)\,dx$ with a sign prediction first.

> **Work:** prediction: on $[0, 2]$, $x - 5$ runs from -5 to -3 — always negative, graph entirely below the axis, so the integral MUST be negative. FTC: $F(x) = x^2/2 - 5x$; $F(2) = 2 - 10 = -8$; $F(0) = 0$; integral $= -8 - 0 = -8$. **Answer: −8 (prediction: negative ✓).** (Classic error: sign slip giving +8 — the prediction step is there to catch it.)

4. Evaluate $\int_0^8 (x - 4)\,dx$ and explain with two triangles.

> **Work:** $F(x) = x^2/2 - 4x$; $F(8) = 32 - 32 = 0$; $F(0) = 0$; integral $= 0$. Explanation: the line crosses the axis at $x = 4$. Below-axis triangle on $[0, 4]$: size $\frac{1}{2} \times 4 \times 4 = 8$, counted -8. Above-axis triangle on $[4, 8]$: size 8, counted $+8$. Signed sum $-8 + 8 = 0$. **Answer: 0 — equal triangles below and above the axis cancel exactly.** (Watch for students who answer "16": that's the TOTAL, not the integral.)

5. True/false: (a) a region's area can be negative; (b) a definite integral can be negative.

> **Work:** (a) **False** — the size of a shape is never negative; no rug has −8 square feet. (b) **True** — the integral is signed accumulation, and when the graph is below the axis (a loss), the integral comes out negative. The minus sign is information (backward/lost/colder), not geometry. **Answer: (a) false (b) true, with the size-vs-signed-accumulation distinction stated.**

6. $v(t) = 6 - 3t$ on $[0, 4]$: net displacement and total distance.

> **Work:** crossing: $6 - 3t = 0$ at $t = 2$. $F(t) = 6t - 3t^2/2$ (check: derivative is $6 - 3t$ ✓). Net = $F(4) - F(0) = (24 - 24) - 0 = 0$. Pieces: $F(2) - F(0) = (12 - 6) - 0 = +6$; $F(4) - F(2) = 0 - 6 = -6$. Total = $|6| + |-6| = 12$. **Answer: net 0 miles (ends at the start); total 12 miles ridden.** (Classic error: reporting net 0 as "didn't move.")

7. $f(x) = x - 2$ on $[0, 5]$: crossing, pieces, net, total.

> **Work:** crossing at $x = 2$. $F(x) = x^2/2 - 2x$. Piece 1: $F(2) - F(0) = (2 - 4) - 0 = -2$. Piece 2: $F(5) - F(2) = (25/2 - 10) - (-2) = 5/2 + 2 = 9/2$ — note minus a negative is PLUS. Net = $-2 + 9/2 = 5/2$ (check straight through: $F(5) - F(0) = 5/2$ ✓). Total = $|-2| + |9/2| = 2 + 9/2 = 13/2$. **Answer: crossing $x = 2$; pieces -2 and $9/2$; net $5/2$; total $13/2$.** (Classic error in piece 2: writing $5/2 - 2 = 1/2$ instead of $5/2 - (-2) = 9/2$.)

8. Pool drains at 6 L/min for 4 min, then refills at 5 L/min for 2 min: net and total.

> **Work:** draining piece: $(-6) \times 4 = -24$ L. Refilling piece: $(+5) \times 2 = +10$ L. (a) Net = $-24 + 10 = -14$: the pool ends with 14 liters LESS. (b) Total moved = $|-24| + |+10| = 24 + 10 = 34$ liters. **Answer: (a) −14 L, i.e. 14 liters less than before; (b) 34 liters moved.**

9. $r(t) = t^2 - 9$ on $[0, 3]$: evaluate and interpret, including why the sign was guaranteed.

> **Work:** $F(t) = t^3/3 - 9t$ (check: derivative is $t^2 - 9$ ✓). $F(3) = 27/3 - 27 = 9 - 27 = -18$; $F(0) = 0$. Integral = -18. Sign was guaranteed: on $[0, 3]$, $t^2 \le 9$, so $t^2 - 9 \le 0$ the whole time — the rate never goes positive, water only leaves. **Answer: −18 → the tank loses 18 liters over the 3 minutes.**

10. Find $b > 0$ with $\int_0^b (x - 3)\,dx = 0$, and what is special about that moment.

> **Work:** $F(x) = x^2/2 - 3x$, so the integral = $b^2/2 - 3b$. Set to 0: $b^2/2 - 3b = 0 \to b(b/2 - 3) = 0 \to b = 0$ or $b = 6$; we need $b > 0$, so $b = 6$. Special moment: at $x = 6$ the above-axis triangle (size $½ \times 3 \times 3 = 9/2$ on $[3, 6]$) has grown to exactly match the below-axis triangle (size $9/2$ on $[0, 3]$) — the gain has just finished repaying the loss. **Answer: $b = 6$.** (Accept any equivalent balance explanation; $b = 0$ alone earns no credit since the problem requires $b > 0$.)